```
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ ./scripts/create-cluster.sh
INFO[0000] portmapping '8443:443' targets the loadbalancer: defaulting to [servers:*:proxy agents:*:p
roxy]
INFO[0000] portmapping '8088:80' targets the loadbalancer: defaulting to [servers:*:proxy agents:*:pr
oxy]
INFO[0000] Prep: Network
INFO[0000] Re-using existing network 'capstone-net' (05e5a86d5e992de9a12a1c6742311710037e4948e79c2b0d
69f5b6b1771f994c)
INFO[0000] Created image volume k3d-capstone-images
INFO[0000] Starting new tools node...
INFO[0000] Starting node 'k3d-capstone-tools'
INFO[0001] Creating node 'k3d-capstone-server-0'
INFO[0001] Creating node 'k3d-capstone-agent-0'
INFO[0001] Creating LoadBalancer 'k3d-capstone-serverlb'
INFO[0001] Using the k3d-tools node to gather environment information
INFO[0001] HostIP: using network gateway 172.18.0.1 address
INFO[0001] Starting cluster 'capstone'
INFO[0001] Starting servers...
INFO[0001] Starting node 'k3d-capstone-server-0'
INFO[0007] Starting agents...
INFO[0007] Starting node 'k3d-capstone-agent-0'
INFO[0011] Starting helpers...
INFO[0011] Starting node 'k3d-capstone-serverlb'
INFO[0017] Injecting records for hostAliases (incl. host.k3d.internal) and for 7 network members into
 CoreDNS configmap...
INFO[0019] Cluster 'capstone' created successfully!
INFO[0019] You can now use it like this:
kubectl cluster-info
NAME                     STATUS   ROLES           AGE   VERSION        INTERNAL-IP   EXTERNAL-IP   OS-
IMAGE           KERNEL-VERSION   CONTAINER-RUNTIME
k3d-capstone-agent-0     Ready    <none>          9s    v1.35.5+k3s1   172.18.0.7    <none>        K3s
 v1.35.5+k3s1   7.2.14-arch1-1   containerd://2.2.3-k3s1
k3d-capstone-server-0    Ready    control-plane   12s   v1.35.5+k3s1   172.18.0.6    <none>        K3s
 v1.35.5+k3s1   7.2.14-arch1-1   containerd://2.2.3-k3s1
NAMESPACE     NAME                                      READY   STATUS              RESTARTS   AGE
kube-system   coredns-8db54c48d-x7scw                   0/1     ContainerCreating   0          6s
kube-system   helm-install-traefik-76f8d                0/1     ContainerCreating   0          4s
kube-system   helm-install-traefik-crd-mxzc9            0/1     ContainerCreating   0          4s
kube-system   local-path-provisioner-5d9d9885bc-4ct7r   0/1     ContainerCreating   0          6s
kube-system   metrics-server-786d997795-5fp9h           0/1     ContainerCreating   0          6s
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$
```

```
Tue Nov 10 02:11:56 PM +07 2026

$ kubectl version; k3d version; docker --version
Client Version: v1.37.2
Kustomize Version: v5.8.1
Server Version: v1.35.5+k3s1
Warning: version difference between client (1.37) and server (1.35) exceeds the supported minor versi
on skew of +/-1
k3d version v5.9.0
k3s version v1.35.5-k3s1 (default)
Docker version 29.9.2, build a82e528901
```

```
Tue Nov 10 02:12:15 PM +07 2026

$ kubectl get nodes -o wide | cut -c1-150
NAME                     STATUS   ROLES           AGE   VERSION        INTERNAL-IP   EXTERNAL-IP   OS-
IMAGE           KERNEL-VERSION   CONTAINER-RUNTIM
k3d-capstone-agent-0     Ready    <none>          75s   v1.35.5+k3s1   172.18.0.7    <none>        K3s
 v1.35.5+k3s1   7.2.14-arch1-1   containerd://2.2
k3d-capstone-server-0    Ready    control-plane   78s   v1.35.5+k3s1   172.18.0.6    <none>        K3s
 v1.35.5+k3s1   7.2.14-arch1-1   containerd://2.2
```

```
Tue Nov 10 02:12:25 PM +07 2026

$ cat Dockerfile
FROM nginx:alpine
COPY default.conf /etc/nginx/conf.d/default.conf
COPY index.html /usr/share/nginx/html/index.html
$ docker build --provenance=false -t localhost:5001/capstone-app:v0.1.0 .
[+] Building 7.8s (8/8) FINISHED                                                docker:default
 => [internal] load build definition from Dockerfile                                      0.0s
 => => transferring dockerfile: 153B                                                      0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine                           7.0s
 => [internal] load .dockerignore                                                         0.0s
 => => transferring context: 2B                                                           0.0s
 => [1/3] FROM docker.io/library/nginx:alpine@sha256:713fc2ebfd92e8217db451477182ee91944b123d2  0.0s
 => => resolve docker.io/library/nginx:alpine@sha256:713fc2ebfd92e8217db451477182ee91944b123d2  0.0s
 => [internal] load build context                                                         0.0s
 => => transferring context: 380B                                                         0.0s
 => CACHED [2/3] COPY default.conf /etc/nginx/conf.d/default.conf                         0.0s
 => CACHED [3/3] COPY index.html /usr/share/nginx/html/index.html                         0.0s
 => exporting to image                                                                    0.8s
 => => exporting layers                                                                   0.0s
 => => exporting manifest sha256:5772d161de9804a9ab1eb43680886887b468ddf5bd64a980ad7bb138f82f8  0.0s
 => => exporting config sha256:5287c4812896ddec36063c4329ca2dc663c045f7446e8120cc923d25e54fba9  0.0s
 => => naming to localhost:5001/capstone-app:v0.1.0                                       0.0s
 => => unpacking to localhost:5001/capstone-app:v0.1.0                                    0.7s
$ docker push localhost:5001/capstone-app:v0.1.0
The push refers to repository [localhost:5001/capstone-app]
1abafee971b9: Layer already exists
881fc932939e: Layer already exists
c7dfd64eeb02: Layer already exists
a746bd1dc828: Layer already exists
da90db506dd5: Layer already exists
ee95d88d4407: Layer already exists
e011f124249b: Layer already exists
b9c6ee6a30f8: Layer already exists
69d81d552257: Layer already exists
c6472f28170b: Layer already exists
v0.1.0: digest: sha256:5772d161de9804a9ab1eb43680886887b468ddf5bd64a980ad7bb138f82f83bd size: 2185
$ curl -s http://127.0.0.1:5001/v2/capstone-app/tags/list
{"name":"capstone-app","tags":["v0.1.0"]}
```

```
Tue Nov 10 02:12:47 PM +07 2026

$ kubectl apply -f app.yaml
namespace/capstone created
configmap/capstone-app-config created
deployment.apps/capstone-app created
service/capstone-app created
ingress.networking.k8s.io/capstone-app created
$ kubectl -n capstone rollout status deploy/capstone-app --timeout=120s
Waiting for deployment "capstone-app" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "capstone-app" rollout to finish: 0 of 2 updated replicas are available...
Waiting for deployment "capstone-app" rollout to finish: 1 of 2 updated replicas are available...
deployment "capstone-app" successfully rolled out
```

```
Tue Nov 10 02:13:27 PM +07 2026

$ kubectl -n capstone get all,ingress,configmap
NAME                                 READY   STATUS    RESTARTS   AGE
pod/capstone-app-594cdffdc4-pmzvw    1/1     Running   0          40s
pod/capstone-app-594cdffdc4-zsvc7    1/1     Running   0          40s

NAME                    TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)   AGE
service/capstone-app    ClusterIP   10.43.215.143   <none>        80/TCP    40s

NAME                              READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/capstone-app      2/2     2            2           40s

NAME                                         DESIRED   CURRENT   READY   AGE
replicaset.apps/capstone-app-594cdffdc4      2         2         2       40s

NAME                                      CLASS     HOSTS                ADDRESS   PORTS   AGE
ingress.networking.k8s.io/capstone-app    traefik   capstone.localhost             80      41s

NAME                              DATA   AGE
configmap/capstone-app-config     2      41s
configmap/kube-root-ca.crt        1      41s
```

```
Tue Nov 10 02:13:39 PM +07 2026

$ kubectl -n capstone describe deploy capstone-app | grep -A8 -E 'Image:|Liveness|Readiness|Limits|Re
quests'
    Image:      registry:5000/capstone-app:v0.1.0
    Port:       8080/TCP
    Host Port:  0/TCP
    Limits:
      cpu:     500m
      memory:  512Mi
    Requests:
      cpu:      100m
      memory:   192Mi
    Liveness:   http-get http://:8080/q/health/live delay=10s timeout=1s period=10s successThreshold=
1 failureThreshold=3
    Readiness:  http-get http://:8080/q/health/ready delay=3s timeout=1s period=5s successThreshold=1
 failureThreshold=3
    Environment Variables from:
      capstone-app-config  ConfigMap  Optional: false
    Environment:           <none>
    Mounts:                <none>
  Volumes:                 <none>
  Node-Selectors:          <none>
  Tolerations:             <none>
Conditions:
```

```
Tue Nov 10 02:13:49 PM +07 2026

$ kubectl -n capstone get deploy capstone-app -o wide
NAME          READY   UP-TO-DATE   AVAILABLE   AGE   CONTAINERS   IMAGES
  SELECTOR
capstone-app   2/2     2            2           61s   app          registry:5000/capstone-app:v0.1.0
  app=capstone-app
$ kubectl -n capstone describe pod -l app=capstone-app | grep -E 'Image:|Image ID:'
    Image:          registry:5000/capstone-app:v0.1.0
    Image ID:       registry:5000/capstone-app@sha256:5772d161de9804a9ab1eb43680886887b468ddf5bd64a98
0ad7bb138f82f83bd
    Image:          registry:5000/capstone-app:v0.1.0
    Image ID:       registry:5000/capstone-app@sha256:5772d161de9804a9ab1eb43680886887b468ddf5bd64a98
0ad7bb138f82f83bd
```

```
Tue Nov 10 02:13:58 PM +07 2026

$ curl -i -H 'Host: capstone.localhost' http://127.0.0.1:8088/q/health/ready
curl: (52) Empty reply from server
```

```
Tue Nov 10 02:14:10 PM +07 2026

$ kubectl -n capstone scale deploy capstone-app --replicas=3
deployment.apps/capstone-app scaled
$ kubectl -n capstone get pods -o wide
NAME                           READY   STATUS              RESTARTS   AGE   IP          NODE
           NOMINATED NODE   READINESS GATES
capstone-app-594cdffdc4-km2wx   0/1     ContainerCreating   0          1s    <none>      k3d-capstone
-agent-0    <none>           <none>
capstone-app-594cdffdc4-pmzvw   1/1     Running             0          83s   10.42.0.4   k3d-capstone
-agent-0    <none>           <none>
capstone-app-594cdffdc4-zsvc7   1/1     Running             0          83s   10.42.1.5   k3d-capstone
-server-0   <none>           <none>
$ sleep 12; kubectl -n capstone get pods
NAME                           READY   STATUS    RESTARTS   AGE
capstone-app-594cdffdc4-km2wx   1/1     Running   0          13s
capstone-app-594cdffdc4-pmzvw   1/1     Running   0          95s
capstone-app-594cdffdc4-zsvc7   1/1     Running   0          95s
$ kubectl -n capstone scale deploy capstone-app --replicas=2
deployment.apps/capstone-app scaled
$ sleep 5; kubectl -n capstone get deploy capstone-app
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
capstone-app   2/2     2            2           100s
```

```
Tue Nov 10 02:14:39 PM +07 2026

$ POD=$(kubectl -n capstone get pod -l app=capstone-app -o name | head -1); echo $POD
pod/capstone-app-594cdffdc4-pmzvw
$ kubectl -n capstone delete $POD --wait=false
pod "capstone-app-594cdffdc4-pmzvw" deleted from capstone namespace
$ curl -s -o /dev/null -w 'HTTP %{http_code}\n' -H 'Host: capstone.localhost' http://127.0.0.1:8088/q
/health/ready
HTTP 000
$ kubectl -n capstone get pods
NAME                           READY   STATUS              RESTARTS   AGE
capstone-app-594cdffdc4-ldzkh   0/1     ContainerCreating   0          1s
capstone-app-594cdffdc4-pmzvw   1/1     Terminating         0          112s
capstone-app-594cdffdc4-zsvc7   1/1     Running             0          112s
$ sleep 15; kubectl -n capstone get pods
NAME                           READY   STATUS    RESTARTS   AGE
capstone-app-594cdffdc4-ldzkh   1/1     Running   0          16s
capstone-app-594cdffdc4-zsvc7   1/1     Running   0          2m7s
```

```
Tue Nov 10 02:15:08 PM +07 2026

$ kubectl -n capstone rollout restart deploy/capstone-app
deployment.apps/capstone-app restarted
$ kubectl -n capstone rollout status deploy/capstone-app
Waiting for deployment "capstone-app" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "capstone-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "capstone-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "capstone-app" rollout to finish: 1 out of 2 new replicas have been updated...
Waiting for deployment "capstone-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "capstone-app" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "capstone-app" rollout to finish: 1 old replicas are pending termination...
deployment "capstone-app" successfully rolled out
$ kubectl -n capstone rollout history deploy/capstone-app
deployment.apps/capstone-app
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
```

SCREENSHOT TAMBAHAN:

```
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ date
Tue Nov 10 02:24:40 PM +07 2026
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ kubectl -n kube-system get pods
 | grep -i traefik
helm-install-traefik-76f8d               0/1     Completed   1          13m
helm-install-traefik-crd-mxzc9           0/1     Completed   0          13m
svclb-traefik-b26b5089-cphhb             2/2     Running     0          10m
svclb-traefik-b26b5089-gmdpg             2/2     Running     0          10m
traefik-9bcdbbd9-n8njz                   1/1     Running     0          10m
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ kubectl -n capstone get ingress
NAME          CLASS     HOSTS               ADDRESS                 PORTS   AGE
capstone-app  traefik   capstone.localhost  172.18.0.6,172.18.0.7   80      12m
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ curl -i -H 'Host: capstone.loca
lhost' http://127.0.0.1:8088/q/health/ready
HTTP/1.1 200 OK
Content-Length: 15
Content-Type: application/json
Date: Tue, 10 Nov 2026 07:24:57 GMT
Server: nginx/1.31.8

{"status":"UP"}[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$
```

```
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ date; POD=$(kubectl -n capstone
 get pod -l app=capstone-app -o name | head -1); echo $POD
Tue Nov 10 02:25:23 PM +07 2026
pod/capstone-app-7479958c8c-k6gwd
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ kubectl -n capstone delete $POD
 --wait=false
pod "capstone-app-7479958c8c-k6gwd" deleted from capstone namespace
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ for i in $(seq 1 10); do curl -
s -o /dev/null -w "%{http_code} " -H 'Host: capstone.localhost' http://127.0.0.1:8088/q/health/ready;
 sleep 1; done; echo
kubectl -n capstone get pods
200 200 200 200 200 200 200 200 200 200
NAME                           READY   STATUS    RESTARTS   AGE
capstone-app-7479958c8c-j2qh6   1/1     Running   0          14s
capstone-app-7479958c8c-x5rmm   1/1     Running   0          10m
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$ kubectl -n capstone get pods
NAME                           READY   STATUS    RESTARTS   AGE
capstone-app-7479958c8c-j2qh6   1/1     Running   0          23s
capstone-app-7479958c8c-x5rmm   1/1     Running   0          10m
[deopg@deopg Cloud_Computing_Common_Capstone_Infrastructure_Package]$
```